Soft vs Rigid Gripper Benchmark for Cloth Grasping

A Force-Depth-Tilt Characterization of Parallel-Jaw and Finray Geometries

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Abstract

Robotic grasping of thin, deformable fabric remains an unsolved problem in industrial automation. Two competing gripper geometries are routinely proposed: rigid parallel-jaw fingers, which offer precise force control but a narrow operational tolerance, and compliant finray-effect fingers, which trade force precision for a wider operational envelope. This thesis presents an experimental benchmark of five gripper variants—two parallel-jaw configurations (rigid stock, and stock plus a Shore-90A thermoplastic polyurethane pad) and three finray configurations (18 mm and 26 mm finger thickness, plus a 5° inclined-seat fixture)—mounted on a Franka FR3 collaborative manipulator. Each variant is characterised across a two-dimensional input space of press depth (the distance the gripper fingertip is driven below the detected cloth surface) and tilt angle (the gripper orientation about world x and y axes relative to the surface normal). Stationary contact force in the gripper z-axis is sampled during each grasp and used as the primary outcome variable, with binary grasp success as a secondary outcome.

The benchmark produced 204 trials over 75 sessions, yielding 139 confirmed successful grasps. The rigid parallel-jaw exhibits a narrow ≈ 1 mm working depth range bounded above by a -60 N safety reflex and below by no-contact; the TPU-padded variant widens this to ≈ 2 mm. The finray variants, by contrast, exhibit a broad ≥ 12 mm working range with a soft plateau near -37 N over the depth interval $+2$ to $+8$ mm.

Beyond this expected compliance advantage, two depth-dependent tilt effects

emerged. First, finray contact force is strongly tilt-sensitive at the no-contact boundary but tilt-insensitive in the soft plateau: a $\pm 5°$ tilt produces a 58% contact force reduction at -0.5 mm depth but only 1% at $+10$ mm depth. Second, the finray gripper exhibits a direction-dependent handedness under tilt about its asymmetric (y) axis at boundary depths—a $\pm 2.5°$ inversion reverses the contact force by a factor of $2.36\times$ at -1.0 mm—but this handedness vanishes at safe operating depths. We propose this as an emergent characterisation of the finray geometry: the deeply pressed finger structure wraps the contact surface symmetrically regardless of tilt sign, while at lighter contact the asymmetric ridge profile dominates the force response.

The implication for cloth-handling system design is that compliant grippers achieve directional robustness only when pressed past a depth threshold; control policies that operate near the no-contact boundary for delicate fabric handling must explicitly model the gripper's directional asymmetry.

Abrégé

[TODO: Résumé en français — requis par McGill GPS. Traduire le résumé anglais; le Département de linguistique ou un collègue francophone peut vérifier la traduction.]

Contribution of Authors

This thesis is the original work of the author, **[TODO: Author Name]**, under the supervision of **[TODO: Supervisor Name]**. The author carried out all experimental design, hardware integration, software development, data collection, analysis, and writing.

The Franka FR3 manipulator and its ROS 2 control stack were maintained by **[TODO: lab name / colleague name]**. The 3D-printed finray fingers were fabricated using lab resources at **[TODO: lab name]**. No portion of this thesis has been previously published.

[TODO: If any chapters or sections derive from joint work or were drawn from a manuscript-based thesis, declare collaborator contributions here per McGill GPS guidelines.]

Acknowledgements

[TODO: Acknowledgements: thesis supervisor, lab members, funding sources, ROS 2 / Franka community, family.]

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Introduction

1.1 Motivation

Fabric handling automation is a long-standing open problem in industrial robotics, with applications ranging from apparel manufacturing and laundry processing to packaging of textile components. Cloth is uniquely difficult to grasp robustly: it deforms continuously under contact, exhibits no fixed graspable feature, and has a near-zero bending stiffness that allows it to slip out of a poorly placed gripper. Conventional rigid-jaw grippers, designed for sturdy objects with well-defined geometry, perform poorly on cloth without elaborate tactile sensing or human-tuned approach trajectories.

A class of compliant grippers based on the finray effect—a passive bending mechanism inspired by fish-fin anatomy—has been proposed as a solution. Finray fingers passively wrap around the contact surface, increasing contact area and gripping force without requiring active force control. This compliance, however, comes at the cost of force-modelling difficulty: the finger geometry deforms under load, the contact point shifts, and tilt-induced asymmetries arise that have no analogue in the rigid-jaw case.

The central question of this thesis is: *to what extent does the finray's compliance trade single-axis force precision for multi-axis operational robustness?* Specifically, how does the finray's contact force respond to deviations in the two design degrees of freedom most likely to be uncontrolled in real deployment: press depth and gripper tilt angle?

1.2 Research Question and Contributions

This thesis answers the following research questions through a controlled experimental benchmark on a Franka FR3 collaborative robot:

1. How does stationary contact force vary with press depth for each of five candidate gripper geometries?
2. For each geometry, what is the maximum tilt angle (about each orthogonal axis) at which a reliable grasp is still produced?
3. Is the dependence of contact force on tilt itself depth-dependent?
4. Does the finray gripper exhibit a direction-dependent (handed) response under tilt about its geometrically asymmetric axis?

The contributions of this work are:

- A reproducible benchmark protocol for cloth-grasp force characterisation on a Franka FR3, with full source code published alongside this thesis.
- A two-dimensional depth–tilt experimental matrix sampling five commercially relevant gripper geometries.
- Empirical evidence of a depth-dependent tilt robustness transition in finray grippers, with quantitative characterisation of the transition depth.
- A documented calibration procedure for per-variant tool-centre-point (TCP) offset using differential force touch-off against a rigid reference.
- A geometric verification framework that disentangles control-frame positioning error from physical gripper compliance under load.

1.3 Thesis Organisation

Chapter 2 reviews the literature on cloth grasping, compliant grippers, and force-controlled approach strategies. Chapter 3 describes the experimental hardware, the kinematic calibration procedure, and the data-acquisition pipeline. Chapter 4 presents the depth-sweep characterisation (Experiment 1) across all five gripper variants at zero tilt. Chapter 4 presents the tilt-sweep characterisation (Experiment 2) at boundary and safe depths. Chapter 6 synthesises the findings into a model of depth-dependent tilt robustness. Chapter 7 concludes and identifies directions for future work.

2

Background and Related Work

2.1 Cloth Grasping in Robotics

[TODO: Literature review on cloth manipulation. Key references to include: Maitin-Shepard et al. (folding clothes; PR2); Kita, Saito (cloth handling for laundry); Yang, Saxena (visual servoing for cloth); Tirumala (cloth perception); recent work from McGill ADL-ROS lab on cloth manipulation. Roughly 4–6 pages.]

2.2 Compliant Gripper Designs

[TODO: Review of compliant gripper literature: Festo Finray fingers (original Fish Fin Effect; Compact); Soft Robotics Inc. pneumatic grippers; Suzuki-Suzumori jamming grippers; Crooks et al. (3D-printed compliant fingers); finger durometer / blade-thickness studies. Roughly 3–4 pages, including geometric description and standard mounting configurations.]

2.3 Force-Controlled Grasping

[TODO: Review of force feedback and impedance-controlled grasping: Hogan's impedance control framework; Franka Cartesian impedance / pose controller specifics; tactile sensing alternatives (FlexiForce A301 / GelSight);

touch-off / probe-down strategies. Roughly 2–3 pages.]

2.4 The Franka FR3 Platform

[TODO: Brief description of the Franka Emika FR3 (formerly Panda) used in this work: 7-DOF, joint torque sensing, libfranka, FCI interface. Cite Franka documentation and any reproducibility studies on the platform. 1–2 pages.]

2.5 Position and Summary

This thesis sits at the intersection of three threads: cloth-handling system design, finray-effect gripper characterisation, and force-controlled manipulation on collaborative robots. Prior work has independently established each thread; the contribution here is a unified experimental treatment of all three on a single platform with consistent instrumentation, enabling direct quantitative comparison between rigid and compliant gripper response under deliberately perturbed conditions.

3

Experimental Setup and Methodology

3.1 Hardware Configuration

3.1.1 Robot Manipulator

A Franka Emika FR3 7-DOF cobot served as the manipulator throughout this work. The arm was operated through the Franka Control Interface (FCI) using `libfranka 0.18.1` and the ROS 2 Humble control stack. Cartesian pose tracking was performed by a custom `fr3_pose_controller` maintained by the McGill Applied Dynamics Lab at `https://github.com/McGill-Applied-Dynamics-Lab/adl-ros2`, parameterised in `config/controllers/fr3_pose/default.yaml`. The controller accepts a sequence of (pose, twist) waypoints with associated time stamps and interpolates between them using a quintic polynomial; for the in-place re-orientation and the final precision descent phases, a cosine half-cycle S-curve interpolator (the so-called "sine descent" implemented in `run_trial.py::_safe_traj`) is used to ensure zero start- and end-state velocities.

Table 3.1: Six gripper variants benchmarked. All finray variants share the same 3D-printed inclined-seat mount; only the seat angle changes between the +2.5° baseline and the +5.0° variant. The parallel-jaw variants do not use an inclined seat.

Variant	Type	Key parameter	Seat angle
`parallel_jaw_stock`	Rigid two-finger (Franka Hand)	—	—
`parallel_jaw_TPU`	Rigid two-finger + TPU 90A pad	2.0 mm pad thickness	—
`finray_18`	Finray-effect 3D-printed	18 mm finger thickness	+2.5°
`finray_18_model2`	Second print of `finray_18`	18 mm finger thickness	+2.5°
`finray_26`	Finray-effect 3D-printed	26 mm finger thickness	+2.5°
`finray_26_5deg`	Finray-26 on 5°-seat mount	26 mm finger thickness	+5.0°

3.1.2 Gripper Variants Tested

The finray fingers and the parallel-jaw compliant pad were 3D-printed from Bambu Lab TPU 90A (Shore-A hardness 90) on a Bambu Lab X1 Carbon printer using the vendor-default TPU 90A profile in Bambu Studio (0.4 mm nozzle, default layer height, default infill, PEI-textured build plate).

Baseline inclined-seat mount. All finray variants are mounted on the Franka Hand via an inclined seat that introduces a fixed +2.5° tilt about the gripper y axis (the gripping direction). The seat is a 3D-printed adapter and is present in *every* variant tested in this thesis unless otherwise noted. Variants whose name contains `_5deg` substitute a +5.0° seat in place of the default +2.5° seat; all other variants (`finray_18`, `finray_18_model2`, `finray_26`, and the parallel-jaw variants) inherit the +2.5° baseline. The +2.5° baseline seat angle was chosen so that the two opposing finray fingers converge inward toward each other rather than splaying apart, closing the residual slit that would otherwise remain between the finger tips when the gripper is fully closed. This ensures firm contact between the fingers along their full length during a successful grasp. Figure [TODO: ref] shows the mounting stack (flange → seat → finger) and defines the seat-tilt convention.

Consequence for reported "0°" cells. Cells reported in Chapter 5 as 0° tilt refer to *zero commanded runtime tilt*; the physical gripper is still inclined at the baseline seat angle relative to the flange. All runtime tilt values (e.g. $\pm 2.5^\circ$ about x or y) are applied *on top of* the baseline seat tilt.

3.1.3 Cloth Specimen

[TODO: insert photograph of the cloth specimen and short caption describing material, thickness, weave, and dimensions.]

3.1.4 Test Surface

[TODO: insert photograph of the test bench in situ and short caption describing the table material, flatness, and cloth-edge orientation relative to the gripper y axis.]

3.2 Reference Frame and Tool-Centre Calibration

3.2.1 Frame Convention

Three frames are relevant: the world (robot base) frame $\mathcal{W}$, the gripper flange frame $\mathcal{F}$, and the gripper tip frame $\mathcal{T}$. The Franka URDF defines $\mathcal{F}$ at the mechanical hand mounting flange; the Franka API `robot.end_effector_pose` reports the pose of $\mathcal{F}$ in $\mathcal{W}$. The tip frame $\mathcal{T}$ is located along the local $+z$ axis of $\mathcal{F}$ by a per-variant offset z_{TCP}, which depends on the physical gripper (finger length, mounting fixtures, pads).

For a gripper at orientation $R \in SO(3)$, the world-frame relationship is

$$\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} + R \begin{bmatrix} 0 \\ 0 \\ z_{\mathrm{TCP}} \end{bmatrix}. \tag{3.1}$$

For the rigid-down-pointing nominal orientation $R_0 = R_x(180^\circ)$, the tip lies directly below the flange and $\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} - z_{\mathrm{TCP}}\hat{\mathbf{z}}$.

The gripper-frame convention adopted here is: the gripping axis (along which the fingers close) is $\mathcal{F}$'s local y axis; the gripper is mirror-symmetric about the xz

plane; consequently, rotation of the gripper about the local y axis is rotation about the geometrically asymmetric axis, while rotation about x is rotation about the symmetric axis.

3.2.2 TCP Offset Calibration

The z_{TCP} value differs by variant because different fingers attach to the same flange. To calibrate, each variant was driven to a common probe position (x_0, y_0) on a known rigid table and a force-feedback touch-off was performed: the gripper descended slowly in z until the external force in z exceeded a $-3\,\mathrm{N}$ threshold (debounced for three consecutive $50\,\mathrm{Hz}$ samples). The flange z at the latch moment was recorded as $z_{\mathrm{tcp}}^{\mathrm{measured}}$.

Taking the rigid `parallel_jaw_stock` variant as the URDF reference ($z_{\mathrm{TCP}} \equiv 0$ by definition), the per-variant TCP offset is

$$z_{\mathrm{TCP}}^{(v)} = z_{\mathrm{tcp}}^{(v),\mathrm{measured}} - z_{\mathrm{tcp}}^{\mathrm{stock},\mathrm{measured}}. \tag{3.2}$$

Table 3.2 summarises the calibrated values used throughout this work.

Table 3.2: Calibrated per-variant TCP offsets from differential touch-off against the stock parallel jaw.

Variant	$z_{\mathrm{tcp}}^{\mathrm{measured}}$ (mm)	$z_{\mathrm{TCP}}^{(v)}$ (mm)
`parallel_jaw_stock`	37.871	0.000
`parallel_jaw_TPU`	39.916	2.044
`finray_18`	131.842	93.971
`finray_26`	132.782	94.911
`finray_26_5deg`	130.094	92.223

The reproducibility of the touch-off was measured as the across-tap standard deviation σ_z in mm. For `finray_26_5deg`, $\sigma_z = 183\,\mu\mathrm{m}$ owing to a wobble component introduced by the inclined seat; the other four variants had $\sigma_z \leq 102\,\mu\mathrm{m}$.

3.3 Force Measurement

External forces $\mathbf{F}_{\text{ext}}$ at the flange were estimated by the Franka internal model from joint torques, available via the `libfranka` RobotState API at 1 kHz and subsampled to 30 Hz for session logging. A free-air bias capture was performed at the start of every session: the gripper was held at a known pose 30–80 mm above the cloth surface and 30 samples were averaged over 1 second to provide F_z^{bias}. This bias was recorded in each session JSON but *not* subtracted from the raw force time series at acquisition time; downstream analyses subtract it where appropriate.

3.4 Tilt-Aware Pose Targeting

When the gripper is tilted, the flange position required to place the tip at a fixed probe location $(x_0, y_0, z_{\text{surf}}^{\text{tip}})$ shifts laterally. From equation (3.1), the required flange position is

$$\mathbf{p}_{\mathcal{F}}^{*} = \mathbf{p}_{\mathcal{T}}^{\text{target}} - R \begin{bmatrix} 0 \\ 0 \\ z_{\text{TCP}} \end{bmatrix}. \tag{3.3}$$

This relation is implemented in the trial controller `run_trial.py`: the target orientation $R = R_{\text{tilt}} \circ R_0$ is constructed at session start by composing the requested tilt rotation $R_{\text{tilt}} = R_y(\theta_y) R_x(\theta_x)$ (extrinsic, world frame) onto the base orientation $R_0 = R_x(180°)$, and the flange position target is updated accordingly. The orientation is then held constant through all subsequent descent and grasp phases, so the tip XY remains at (x_0, y_0) throughout the trial.

For a $+2.5°$ tilt about x on the `finray_26` variant ($z_{\text{TCP}} = 94.911\,\text{mm}$), the predicted flange offset is

$$\Delta y_{\mathcal{F}} = -z_{\text{TCP}} \sin(2.5°) \approx -4.14\,\text{mm}. \tag{3.4}$$

Chapter 4 reports the empirical lateral offsets measured under load and the residuals from this geometric prediction, which we attribute to finger compliance under press.

3.5 Data Pipeline

3.5.1 Per-Trial Recording

Each trial directory `trial_NNN/` contains:

- `config.yaml` — snapshot of the variant config at trial time;
- `run_config.json` — depth, tilt, ground-truth source, speeds, gripper mode;
- `pose_wrench.csv` — high-rate (30 Hz) actual + commanded pose, RPY, wrench, gripper width per row, tagged with the current trial stage.

3.5.2 Per-Session Recording

Each session directory `depth_*/` contains in addition:

- `depth_summary.json` — header metadata + array of per-trial verdicts (`success`, `fail_reason`); operator notes;
- `ground_truth_used.json` — snapshot of the GT JSON loaded;
- `session_log_*.npz` — numpy archive of the 30 Hz background sample stream, with stage labels and tap indices.

3.5.3 Verdict Recording and Outcome Classification

Trial outcomes were classified into three categories rather than a binary success/failure. After each trial the operator was prompted to distinguish *success* (the gripper held the cloth throughout the lift stage) from *no-grasp* (the trial executed to completion and the stationary force sample was valid, but the cloth slipped from the gripper during or after the lift). Trials in which the FCI control daemon aborted the motion mid-trial were recorded as *failed trials* and excluded from all aggregate analyses; the two most common triggers were:

- the Franka safety reflex firing on excessive predicted joint force (`auto_failed_excess_force` which typically occurred when the parallel-jaw variants were commanded to depths beyond their narrow force-limited working range;

- the Franka safety reflex firing on acceleration discontinuity between two consecutive Cartesian trajectory segments, which was observed intermittently during the chunked touch-off descent and mitigated through the trajectory-emission-floor modifications documented in Section 3.2.2.

Both trigger classes leave the trial's `success` field as `null` in the per-trial JSON; no-grasp trials, by contrast, carry `success = false` with a valid force record and are retained for engagement-force analysis.

3.6 Trial Procedure

A single trial proceeds:

1. Open gripper (Franka Hand or finray release width);
2. Bulk descent: smooth multi-waypoint trajectory from the lifted pose down to a fine clearance (+5 mm above surface), filtered to exclude any upward leg (preventing direction reversals at zero-twist interior waypoints);
3. Settle: 2 s pause at the fine-clearance pose;
4. Final approach: cosine-S-curve descent from the fine clearance to the press target $z_{\text{surf}} - d$, where d is the trial press depth;
5. Grasp: force-controlled close (Franka grasp action at 80 N or 20 N nominal, with ± 100 mm ϵ to disable the position-acceptance check that would otherwise cause auto-release on a sub-spec final width);
6. Settle: 1 s pause;
7. Stationary sample: 30 samples of $\mathbf{F}_{\text{ext}}$ captured over 1 s during the `grasp_stationary_sampl` stage;
8. Lift to $\sim +80$ mm above surface for visual inspection;
9. Operator verdict prompt;
10. Open gripper and reset for next trial.

3.7 Statistical Treatment

For each (variant, depth, tilt) cell, the contact force is reported as

$$\bar{F}_z = \frac{1}{N}\sum_{i=1}^{N} F_z^{(i)}, \qquad \sigma_{F_z} = \sqrt{\frac{1}{N}\sum (F_z^{(i)} - \bar{F}_z)^2}, \tag{3.5}$$

where N is the total number of stationary samples pooled across all clean trials in the cell. Trials with null verdict (FCI crash) are excluded. Trial-to-trial variance is the dominant source of dispersion; the within-trial sample variance ($\sim 0.1\,\mathrm{N}$) is small compared to it.

4

Experimental Design and Procedure

The empirical campaign is organised into two sequential phases against a single experimental apparatus. Phase 1 establishes the contact-force versus press-depth relationship for each gripper variant at nominal zero-tilt orientation, and phase 2 perturbs the same measurement with $\pm 2.5°$ tilts about the two orthogonal Cartesian axes. Together the two phases define a three-dimensional (depth, tilt-axis, tilt-sign) cell grid whose per-cell measurements constitute the data corpus reported in Chapter 5.

4.1 Study Design

The gripper roster comprises six variants organised into two families:

- **Finray family** (four variants): `finray_18`, `finray_18_model2` (second print of the same CAD), `finray_26` (larger blade), and `finray_26_5deg` (`finray_26` on a 5°-inclined seat).
- **Parallel-jaw family** (two variants): `parallel_jaw_stock` (bare rigid jaws) and `parallel_jaw_TPU` (rigid jaws with a 2 mm TPU 90A compliant pad).

The same fabric specimen (single-layer cotton) rests on the same acrylic table surface throughout the campaign. The Franka Emika Panda arm holds the gripper in a fixed base orientation $R_0 = R_x(180°)$ for phase 1, and applies signed rotations $R_{\alpha,\beta} = R_x(180°)R_x(\alpha)R_y(\beta)$ for phase 2, with $(\alpha, \beta) \in \{(\pm 2.5°, 0), (0, \pm 2.5°)\}$.

4.2 Depth Grid

The depth axis is defined relative to the per-session detected cloth surface via the `ground_truth_finder` touch-off procedure described in Chapter 3. The depth grid was chosen adaptively per gripper family:

- For the rigid parallel jaws, depths are spaced at 0.25–0.5 mm intervals over -1 to $+4$ mm, close to the narrow force-limited working range.
- For the finrays, depths are spaced at 0.5–1.0 mm intervals over -2 to $+16$ mm, covering the four regimes (boundary, knee, valley, recovery) identified in Chapter 5.

Per (variant, depth, tilt) cell, three trials were collected. Boundary depths were routinely retested when within-cell standard deviation exceeded 1 N, up to a maximum of two additional cells per (depth, tilt) combination.

4.3 Tilt-Aware Pose Geometry: Verification

Before reporting force results, the implementation of the tilt-aware tool-centre-point (TCP) tracking equation was verified empirically. Table 4.1 shows the measured flange XY shift during the stationary grasp sample versus the geometric prediction for `finray_26` at four representative (depth, tilt) cells.

At boundary depth, the residual is ≤ 0.21 mm—essentially the lateral positioning noise of the arm under light load. At safe depth, both positive and negative tilts exhibit a $+1.2$ mm common-mode offset in the same $+x$ direction, regardless of tilt sign. We interpret this common-mode offset as evidence of finger compliance under load: the ~ 36 N axial press deflects the soft finray blade laterally by approximately 1.2 mm in a structurally preferred direction. The differential offset (between $+y$ and $-y$ tilts) remains correctly predicted by the geometric model in both depth conditions.

Table 4.1: Measured vs. predicted flange-XY offset during the stationary grasp sample at four cells. Predicted offset uses $z_{\mathrm{TCP}} = 94.911$ mm.

Cell	Tilt cmd	Predicted $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Measured $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Residual	Mean $\lvert\mathbf{F}\rvert$
-1.0 mm	$+2.5°$ y	$(+4.14, +0.00)$	$(+4.35, -0.03)$	0.21 mm	-8.00 N
-1.0 mm	$-2.5°$ y	$(-4.14, +0.00)$	$(-4.02, -0.04)$	0.13 mm	-3.39 N
$+10.0$ mm	$+2.5°$ y	$(+4.14, +0.00)$	$(+5.35, -0.28)$	1.25 mm	-36.26 N
$+10.0$ mm	$-2.5°$ y	$(-4.14, +0.00)$	$(-2.93, -0.29)$	1.21 mm	-35.95 N

This verification confirms that the control-frame implementation of the tilt-aware TCP tracker is correct, and any asymmetries observed in the contact force reflect physical gripper behaviour rather than control-system errors.

4.4 Rotation Tracking Verification

A natural concern is whether the Franka arm faithfully realises the commanded tilt orientation. The 30 Hz session log records actual and commanded RPY at every sample, enabling a direct measurement of orientation tracking error

$$\delta(t) = \|\mathrm{RPY}_{\mathrm{actual}}(t) - \mathrm{RPY}_{\mathrm{cmd}}(t)\|_2. \tag{4.1}$$

Across five representative sessions, the mean steady-state δ was 0.05° with a worst-case 0.13°. Transient excursions up to $\sim 5°$ occur exclusively within the `orientation_lock` stage at session start, before any descent or grasp, and are filtered out of trial-level analyses.

A 2.5° commanded tilt is therefore 50× the noise floor of the arm's rotation tracking; the observed force differences cannot be attributed to arm rotation error.

4.5 Trial Protocol and Outcome Classification

Each trial follows a fixed six-stage sequence: (1) set the commanded pose above the target, (2) descend to the pre-contact height, (3) contact and press to the target depth, (4) grasp with the gripper's default close command, (5) lift by a fixed clearance,

(6) operator confirms whether the cloth is held. A stationary-sample window of at least 2 seconds during the grasped-and-pressed stage provides the force measurement.

Trial outcomes are classified into three categories:

Success: the trial executed to completion, contact force was sampled cleanly during the stationary window, and the cloth remained gripped after the lift stage.

No-grasp: the trial executed to completion and contact force was sampled cleanly during the stationary window, but the cloth slipped from the fingers during or after the lift stage. The force measurement remains valid as an *engagement-force* characterisation.

Failed trial: the trial did not execute to completion. The controller either aborted with an excess-force safety event or the operator interrupted before the stationary window completed. The measurement is invalid and the trial is excluded from analyses.

The rationale for this three-way taxonomy—rather than a binary success/failure classification—is discussed in Section 6.5.

5
Results

This chapter reports the empirical findings across all six gripper variants (`finray_18`, `finray_18_model2`, `finray_26`, `finray_26_5deg`, `parallel_jaw_stock`, `parallel_jaw_TPU`) organised around the primary measurement (contact force F_z) at swept command depths and $\pm 2.5°$ tilts about the x and y axes. All values are F_z-bias-corrected using the per-session Franka F_{ext} zero-load reading and pooled across trials with matching command depths and tilts unless otherwise noted. Interpretation is deferred to Chapter 6.

5.1 Canonical Force-Depth Curves at $0°$

Under zero tilt, all four finray variants exhibit a stereotyped *bell-plus-valley* force-depth response with four regions: (i) a boundary regime where contact force rises steeply with press depth, (ii) a knee peak around +2 to +3 mm past the detected surface, (iii) a valley trough near +6 to +8 mm, and (iv) a recovery regime deeper than +10 mm. Table 5.1 summarises the key features.

The four features are within ± 1 mm of the same depth locations across all four variants. The absolute force magnitudes scale approximately with the frontal contact area of the finger.

Table 5.1: Key features of the 0° force-depth curve for each finray variant. Peak $|F_z|$ is the maximum observed contact force in the knee region; valley $|F_z|$ is the local minimum between the knee peak and the deep-regime recovery.

Variant	Knee peak $\|F_z\|$	Peak depth	Valley $\|F_z\|$	Valley depth
`finray_18`	35.4 N	+1.5 mm	31.9 N	+6 mm
`finray_18_model2`	31.2 N	+3.0 mm	25.2 N	+8 mm
`finray_26`	$\sim$ 45 N	+2.0 mm	$\sim$ 41 N	+6 mm
`finray_26_5deg`	$\sim$ 50 N	+2.5 mm	$\sim$ 43 N	+6 mm

5.2 Bipolar Tilt Bias on the Finray Family

A ±2.5° tilt produces a signed bias $b(d,a) = |F_z^{+2.5°}(d,a)| - |F_z^{-2.5°}(d,a)|$ whose depth trajectory itself follows a bipolar shape: a strong signed peak at boundary depths, a zero-crossing near the knee, a sign-reversed plateau through the valley, and depth-gated attenuation toward zero at deep press. The clearest example is the y-axis bias on `finray_26_5deg`, which has full coverage across 14 sign-flip pairs from −2 to +16 mm:

- Boundary peak: $b = +5.40 \pm 0.12$ N at $d = -1.5$ mm.
- Zero crossing: between $d = -0.5$ mm ($b = +2.00$ N) and $d = +1.5$ mm ($b = -1.35$ N).
- Knee plateau: $b = -1.57 \pm 0.21$ N at $d = +2.0$ mm, persisting to $d = +3.0$ mm.
- Depth-gated recovery: $|b| < 0.9$ N at all depths beyond $d = +4$ mm.

The x-axis bias on `finray_26_5deg` has a similar bipolar shape but with opposite sign at the boundary ($b_x = -4.17 \pm 0.32$ N at $d = -1.5$ mm).

5.3 Cross-Axis Sign Reversal at the Boundary

At boundary depths, the x-axis and y-axis biases have *opposite signs* on `finray_26_5deg` (Table 5.2).

Table 5.2: Bias on `finray_26_5deg` at boundary depths. x-axis and y-axis biases have opposite signs. Values are $|F_z^{+2.5°}| - |F_z^{-2.5°}|$ in newtons.

Depth	x-bias	y-bias	Sign product
-1.5 mm	-4.17 ± 0.32	$+5.40 \pm 0.12$	$-$
-1.0 mm	-3.94 ± 0.18	$+4.27 \pm 0.07$	$-$
-0.5 mm	-3.51 ± 0.39	$+2.00 \pm 0.24$	$-$

The same sign-reversal pattern is present on `finray_18` at $d = -1.0$ mm (x-bias $= -2.44 \pm 0.17$ N; y-bias $= +4.49 \pm 0.47$ N), demonstrating that the effect is not specific to the 5° mount condition.

5.4 Mount-Tilt Alignment: 5° Mount as a Depth Shift

After a two-parameter best-fit alignment (single depth shift plus single force offset), the 5°-mounted variant collapses onto the flat-mounted variant with residual RMSE below 2 N on both tilt axes independently (Table 5.3).

Table 5.3: Best-fit alignment of `finray_26_5deg` onto `finray_26`. The two axes yield independent estimates of the depth shift within 0.03 mm of each other.

Axis	Depth shift	Force offset	Alignment RMSE
x	-1.14 mm	$+2.7$ N	1.75 N
y	-1.12 mm	$+3.1$ N	1.58 N
mean	**$-$1.13 mm**	$+2.9$ N	1.66 N

5.5 Print-to-Print Manufacturing Variability

Two independent prints of the same `finray_18` CAD geometry tested on the same day yield a two-parameter alignment of depth shift $+0.85$ mm and force offset -5.5 N with residual RMSE 1.82 N. The 5.5 N force offset is the canonical same-session print-to-print variability, approximately 15% of the knee peak force.

Cross-session drift on the *same* print, measured by comparing finray_18 data collected on separate days approximately three weeks apart, is approximately 7–10 N at knee and knee-adjacent depths. Cross-session drift on the same specimen is therefore larger than same-session print-to-print variability.

The two prints also differ qualitatively in x-axis bias sign: finray_18 exhibits a bipolar x-bias (negative at boundary, positive at knee), whereas finray_18_model2 exhibits a monotonically negative x-bias across the entire measured range.

5.6 Trial Outcome Distribution

Trial outcomes were classified into three categories: *success* (cloth held after lift), *no-grasp* (trial completed with valid F_z measurement but cloth slipped at lift), and *failed trial* (controller aborted or trial did not complete a stationary window). The failed-trial category was excluded from force-curve analyses; no-grasp trials contribute valid force measurements. No-grasp fractions at each variant's boundary depth are reported in Table 5.4.

Table 5.4: No-grasp fraction at each variant's boundary depth. Cells that produce cleanly measurable contact force but fail to hold the cloth on lift.

Variant	Boundary depth	No-grasp fraction
finray_18	−2.0 mm	3/3
finray_18_model2	−1.0 mm	3/3
finray_26	−0.5 mm	$\sim 40\%$
finray_26_5deg	−2.0 mm	6/6
parallel_jaw_stock	+1.0 mm	4/6
parallel_jaw_TPU	+0.5 mm	varies with tilt

5.7 Working-Range and Failure Comparison

The parallel-jaw variants exhibit successful-grasp windows of approximately 1–2 mm in commanded depth (+1.5 to +3.5 mm on parallel_jaw_stock at 0°; +3.0 to +3.5 mm on parallel_jaw_TPU). The finray variants exhibit successful-grasp windows of 8–12 mm on all four variants (+1 mm to +14 mm and beyond).

The parallel jaws auto-fail on the Franka excess-force safety monitor at depths only 1–1.5 mm beyond their successful-grasp window; the finrays never triggered the safety monitor within the tested −2 to +16 mm range on any variant.

6

Discussion

The results reported in Chapter 5 are interpreted in the same six thematic groups, with an emphasis on the physical mechanisms that produce the observed force behaviour and the implications for cloth-handling system design.

6.1 Bell-Plus-Valley Curve Shape

The bell-plus-valley force-depth shape observed on all four finray variants (Section 5.1) is consistent with the finray finger operating in three sequential contact regimes as press depth increases. In the boundary regime, only the distal tip of the finger contacts the cloth; force rises roughly linearly with penetration because the contact area grows in proportion to displacement. At the knee, the full ridged section of the finger is engaged and the finger resists compression through its cross-web structure, producing the peak force. Past the peak, the ribs partially buckle and the finger folds back on itself; contact area increases but effective stiffness drops, producing the valley. At recovery depths, the finger has fully wrapped and further penetration is resisted by structural compression through the finger base, producing the second rise.

The finding that the four features (boundary knee, peak, valley, recovery) occur at the same command depths (± 1 mm) across all four finray variants indicates that this three-regime response is a geometric property of the finray shape itself and not sensitive to finger thickness (18 vs 26 mm) or to a 5° mount tilt. Force magnitudes scale with contact area, as expected, but the depth locations of the transitions do not.

6.2 Cross-Axis Sign Reversal: Mechanism

The finray finger has *mirror symmetry* about the x-axis (the mirror plane between the two opposing fingers) but not about the y-axis (the gripping direction, which breaks left-right equivalence by construction). The observed y-axis bias at boundary depths is therefore consistent with the intrinsic finger geometry: a $+2.5°$ rotation about the gripping axis loads the two fingers differently and produces a signed force difference.

The x-axis bias is more surprising. If the finger and cloth were both symmetric about the mirror plane, a $\pm 2.5°$ rotation about the mirror axis should produce zero bias by symmetry. That the x-bias is nonzero *and of opposite sign to the y-bias* indicates the symmetry-breaking is not on the finger side. The most parsimonious explanation is that the cloth specimen has a directional weave-warp axis that is not aligned with the mirror plane of the gripper. A $+2.5°$ x-rotation of the gripper then produces a systematic cloth-load asymmetry that couples through the whole-hand contact into a signed force difference. The finding that this x-bias is reproducible across two independently mounted variants (`finray_18` and `finray_26_5deg`) is consistent with a shared external source (cloth or table) rather than a per-print effect.

The prediction from this mechanism is that rotating the cloth specimen 90° in the lab frame should flip the sign of the x-bias while leaving the y-bias approximately unchanged (because the intrinsic finger asymmetry does not depend on the cloth orientation). This is the single most important follow-up experiment to isolate the cloth contribution.

6.3 Mount-Tilt Equivalence: Geometric Interpretation

The observation that a 5° pre-tilt of the finray mount is equivalent to a -1.13 mm depth shift, with axis-independent shift within 0.03 mm on the two axes (Table 5.3), admits a simple geometric interpretation. A $5°$ tilt of a rigid finger of length L produces a projected vertical offset at the contact tip of $L \sin(5°) = 0.087L$. Setting this equal to the observed 1.13 mm shift gives $L_{\text{eff}} \approx 13$ mm. Because the finray finger is physically ~ 50 mm long, this L_{eff} corresponds to a contact-engagement point roughly 25% up the length of the finger from its distal tip, consistent with the load being concentrated on the proximal portion of the finray's curved contact surface rather than at the tip.

The axis-independence of the shift (0.02 mm apart on the two axes) is a strong constraint: it rules out any mechanism in which the pre-tilt produces a different effective load-bearing length on the two axes. The $5°$ mount therefore acts as a strictly rigid translation of the operating point in the depth dimension, independent of the tilt axis applied at runtime. This makes the $5°$ mount a useful design lever: the finray's operating envelope can be shifted a chosen 1.13 mm in depth without altering the underlying force-depth sensitivity, at the cost of the ~ 3 N vertical force offset.

6.4 Manufacturing Variability vs Session Drift

The finding that same-day print-to-print variability (5.5 N) is *smaller* than same-print cross-session drift (7–10 N) has two consequences worth stating explicitly.

First, from a design-decision perspective, print-to-print variability is well within the range where any bias claim below $2\sigma \approx 1$–2 N is unlikely to reproduce across different physical instances of the same CAD. Any cross-variant force-magnitude claim in this thesis with an effect size below approximately 2 N should be regarded as tentative and requires cross-print replication before being carried into product decisions.

Second, from an experimental-methodology perspective, session drift is the dominant noise source in cloth-grasping benchmarks of this kind. Any measurement campaign that pools data across days must either report cross-day drift as a separate variance component or restrict its headline claims to same-session pairs. In this thesis, all bias values are reported for same-session pairs unless explicitly labelled otherwise; cross-day comparisons are used only for shape-consistency observations, not for headline effect sizes.

The qualitative x-axis bias sign flip between `finray_18` and `finray_18_model2` (bipolar vs monotonically negative) is a puzzle that this thesis cannot resolve with the current data. Two prints from the same slicer, same infill, same material, and same printer produced measurably different tilt-response behaviours on the x-axis but similar behaviour on the y-axis. The most likely mechanism is small differences in rib-wall thickness or infill density that manifest as asymmetric buckling behaviour under x-axis load, but verifying this requires post-hoc CT-scanning or a controlled study with more prints. The finding is noted here as an interesting boundary condition on manufacturing-invariance claims for compliant grippers.

6.5 Failure-Mode Taxonomy: Why the Distinction Matters

The trial-outcome taxonomy of success, no-grasp, and failed trial (Section 5.6) is deliberately three-valued rather than binary because the three categories map to physically distinct regimes with distinct control implications.

Treating no-grasp trials as failed trials would systematically delete data from the boundary regime, where cloth engagement is intermittent but force measurement is clean. Boundary-regime force values are essential for characterising the low end of the working range, so their exclusion would truncate the observed force-depth curves and mask the boundary-tilt biases that are one of the headline findings.

Treating failed trials as no-grasps in the opposite direction would pollute the force averages with spurious near-zero readings from aborted controllers, inflating apparent variance and eroding the signal-to-noise of the bias estimates. The three-way taxonomy is the minimum classification that separates “the trial produced valid measurement” from “the grasp was successful” from “the trial was invalid.”

For deployment implications, the no-grasp regime is precisely where a force-feedback grasp controller could intervene productively: the contact force is being measured and the depth is known, but the lift is failing. A closed-loop controller monitoring $|F_z|$ during lift could detect the impending slip and either deepen the press or abort before dropping the cloth outside the pick zone. Distinguishing no-grasp from failed-trial in the data record makes this control strategy visible in the offline analysis.

6.6 Deployment-Envelope Implications for Cloth-Handling Systems

The parallel-jaw versus finray comparison (Section 5.7) puts an approximately 8× factor between the two classes in tolerance to depth command noise. This factor is large enough that any cloth-handling system whose depth uncertainty exceeds 1 mm should default to a finray class gripper unless there is a specific force-precision requirement that the finray cannot meet.

The absence of excess-force safety events on any finray variant across the full −2 to +16 mm range indicates that finray-based systems can be deployed open-loop with no depth clipping controller and no safety-limit tuning. Parallel-jaw systems require an active depth governor whose accuracy budget must be at least a factor of two tighter than the successful-grasp window, i.e. sub-half-millimeter command precision, to avoid frequent aborts.

The three-position design envelope proposed in the earlier version of this discussion (safe-plateau finray, boundary finray with tilt control, and rigid jaw with compliant pad) survives the expanded dataset but with a refinement: the compliant-pad variant (`parallel_jaw_TPU`) has narrower successful-grasp coverage than the discussion originally suggested, and its excess-force susceptibility is closer to `parallel_jaw_stock` than to finray. The compliant-pad design remains a useful hybrid but its robustness benefit is qualitative (better boundary behaviour) rather than transforming the parallel jaw into a finray substitute.

6.7 Limitations

Single cloth specimen and single surface. The reported working ranges and cloth-fixture asymmetries were measured on one cotton cloth specimen on one acrylic table surface. The absolute numbers will shift with cloth thickness, weave direction, friction, and surface compliance. The specific cross-axis sign reversal attributed to cloth weave in Section 6.2 requires a 90°-rotated cloth control to isolate the cloth contribution from the fixture contribution.

Cross-session drift. The dominant noise source in this dataset is cross-session drift on the same specimen, exceeding print-to-print manufacturing variability. All same-day comparisons reported here are internally valid; cross-day claims are qualified throughout.

Parallel-jaw x-axis coverage. The parallel-jaw variants were not swept across the x-axis tilt sign; the parallel-jaw bias values are restricted to the y-axis. Whether the x-axis bipolar bias observed on the finrays has an analogue on the rigid grippers is untested.

Boundary-depth sample sizes. Boundary-depth cells frequently have $n = 3$ and inter-trial standard deviations of 0.5–1.5 N driven by cloth-fold state variability that is not further characterised. Boundary values are honest measurements with correspondingly wide error bars.

Success rate as a binary rating. The success/no-grasp distinction is operator-assessed after the lift stage; a finer-grained grip-force-during-lift measurement would resolve marginal cases that the current binary rating collapses.

6.8 Code and Data Availability

All source code, raw trial data, ground-truth JSON files, and analysis scripts are published in the project repository at **[TODO: public GitHub URL]**. The thesis was written against commit **[TODO: commit hash]**.

7

Conclusion and Future Work

7.1 Summary of Contributions

This thesis presented a controlled benchmark of five gripper variants on a single platform (Franka FR3), measuring stationary contact force as a function of press depth and tilt angle. The principal findings are:

1. Rigid parallel-jaw grippers exhibit narrow ($\sim$ 1–2 mm) working depth ranges with steep force-depth slopes (-19 to -26 N/mm).
2. Finray-effect grippers exhibit broad (10–12 mm) working ranges with a flat soft plateau (-1.2 N/mm in the linear regime).
3. Finray contact force is strongly tilt-sensitive at the no-contact boundary (53% reduction at 5° tilt for the boundary depth of -0.5 mm).
4. Finray contact force is tilt-insensitive in the soft plateau ($\leq$ 1% change at 5° tilt for the safe depth of $+10$ mm).
5. Finray response is direction-dependent under tilt about the asymmetric (y) axis at boundary depths (2.4$\times$ ratio between $\pm$2.5°), but symmetric in the soft plateau.

The depth-dependent tilt robustness identified here had not been quantitatively characterised in prior literature, to the author's knowledge, and constitutes the principal empirical contribution.

7.2 Future Work

Cross-variant tilt sweep. The natural extension is to repeat Chapter 4 on the remaining four variants. The rigid parallel-jaw case is of particular interest as a hypothesised tilt-invariant baseline.

Tilt-sensitivity model. An analytical model relating finger curvature, contact compliance, and tilt-induced contact-point migration would connect the present empirical findings to the broader continuum-mechanics literature on soft contact.

Multi-fabric study. All trials used a single cloth specimen. Replicating the matrix on fabrics of varying thickness, weave, and surface treatment would establish the applicability range of the depth-tilt-robustness curves identified here.

Closed-loop control. The boundary-depth regime, where finray directional asymmetry dominates, motivates a tilt-compensating controller using inline force feedback. Implementing such a controller and validating it against the open-loop boundary baseline would be the natural follow-on for a system-design thesis.

Sim2real validation. A simulation of the finray finger in a compliant-contact physics engine could be calibrated against the empirical force-depth-tilt surface from this thesis, providing a digital twin for the gripper variants.

List of Publications

Published:

- My Papers

Bibliography

Acronyms

MOG	Multiplayer Online Games
NAT	Network Address Translation